AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2019

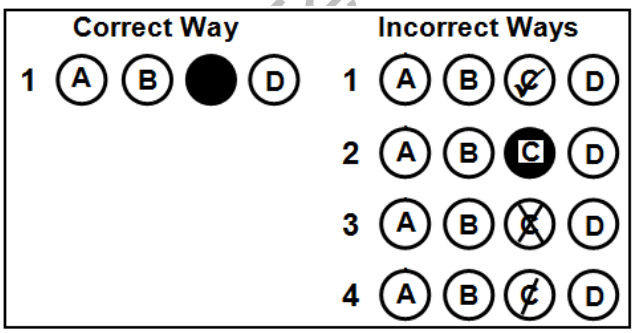
Chemistry Paper I

Time: 50 minutes Marks: 35

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 35 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. How much volume is occupied by 5.2 g of chlorine gas at standard temperature and pressure (STP)?

   (**Note**: Molar mass of Cl = 35.5 g mol^{-1})

   A. 0.164 dm^3
   B. 0.328 dm^3
   C. 1.641 dm^3
   D. 3.283 dm^3

2. In a laboratory, a student decomposes 10.5 g of $CaCO_3$ to obtain CaO and CO_2.

   If 5.25 g of CaO is produced in this process, what is the percentage yield of this reaction?

   (**Note**: Molar mass of Ca = 40 g mol^{-1}, C = 12 g mol^{-1} and O = 16 g mol^{-1})

   A. 50%
   B. 89.29%
   C. 112%
   D. 200%

3. Which of the following sets of quantum numbers is permissible?

	n	***l***	**m**	**s**
A	2	1	+1	+1
B	3	2	−1	−1
C	4	3	+2	+½
D	3	1	−2	−½

4. According to Hund's rule, the electronic configuration of $^{52}_{24}Cr$ in ground state is

   A. $[Ar]4s^0, 3d^6$
   B. $[Ar]4s^2, 3d^4$
   C. $[Ar]4s^1, 3d^1_{xy}, 3d^1_{yz}, 3d^1_{xz}, 3d^1_{x^2-y^2}, 3d^1_{z^2}$
   D. $[Ar]4s^2, 3d^1_{xy}, 3d^1_{yz}, 3d^1_{xz}, 3d^1_{x^2-y^2}, 3d^0_{z^2}$

5. The radius of the 2nd orbit of hydrogen atom is

   (**Note**: $a° = \frac{\in_\circ h^2}{\pi m e^2} = 0.529\ A°$)

   A. 0.265 A^o
   B. 1.058 A^o
   C. 2.116 A^o
   D. 3.781 A^o

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6. $Al^{3+}_{(aq)} + 6H_2O_{(l)} \leftrightharpoons Al(H_2O)^{3+}_{6(aq)}$

In the given reaction, the reactant aluminium ion can be classified as a/ an

A. Lewis acid.
B. Lewis base.
C. neutral species.
D. amphoteric species.

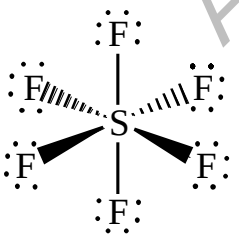
7. If a solution has pH 5, its hydroxide ion and hydronium ion concentration will be

	Hydroxide (OH^-) Ion	**Hydronium (H_3O^+) Ion**
A	10^{-5}	10^{-9}
B	10^{-5}	10^{-5}
C	10^{-9}	10^{-9}
D	10^{-9}	10^{-5}

8. Inability of any solvent to differentiate among the relative strength of all acids is termed as

A. basicity.
B. solubility.
C. buffer action.
D. levelling effect.

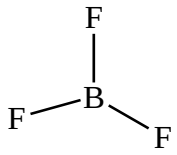
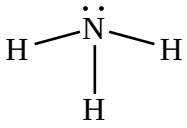
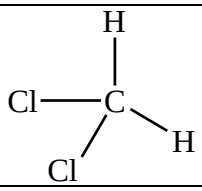
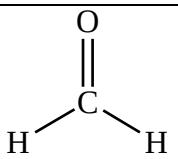
9. The molecular geometry of sulphur hexafluoride (SF_6) is



A. octahedral.
B. square planar.
C. trigonal pyramidal.
D. trigonal bipyramidal.

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10. Which of the following molecules has a net dipole moment of zero?

	
A	B
	
C	D

11. In the formation of a molecule of ethyne,

A. each carbon atom forms three hybrid orbitals.
B. two sigma bonds are formed between the two carbon atoms.
C. each carbon atom undergoes sp-sp overlap with another carbon atom.
D. overlap of unhybridised p-orbitals forms one pi bond between the two carbon atoms.

12. How many lone pairs and bond pairs of electrons are present around the central atom in a molecule of SO_4^{2-}?

	Lone Pairs of Electrons	**Bond Pairs of Electrons**
A	2	2
B	2	4
C	0	4
D	1	3

13. How much pressure in Nm^{-2} is exerted by one mole of a gas enclosed in a vessel of volume 59.4 m^3 at 120°C? (**Note**: R = 8.314 $JK^{-1}mol^{-1}$)

A. 16.79 Nm^{-2}
B. 21.41 Nm^{-2}
C. 38.21 Nm^{-2}
D. 55.01 Nm^{-2}

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14. The set of conditions that is suitable for a gas to be non-ideal is

	Temperature	Pressure
A	High	High
B	Low	High
C	Low	Low
D	High	Low

15. The least volatile organic solvent at room temperature is

A. ether.
B. acetone.
C. acetic acid.
D. dichloromethane.

16. The process of diffusion is visible through the

A. disappearance of water from a pool in hot summers.
B. appearance of fog on the mirror when taking hot shower.
C. appearance of brown colour of coffee granules in a cup of hot milk.
D. disappearance of dry ice creating a smoky effect at room temperature.

17. An example of amorphous solids is

A. graphite.
B. table salt.
C. paraffin wax.
D. rhombic sulphur.

18. Calcium carbonate shows the phenomenon of polymorphism. The polymorphs of calcium carbonate

A. show same physical properties.
B. exist in the same crystalline form.
C. show different chemical properties.
D. exist in more than one crystalline form.

19.

- **Axes**: $a \neq b \neq c$
- **Angles**: $\alpha = \gamma = 90°$, $\beta \neq 90°$

The crystal system that is represented by the given unit cell dimensions is

A. triclinic system.
B. rhombic system.
C. hexagonal system.
D. monoclinic system.

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20. The balanced chemical equation for the given K_c expression would be

$$K_c = \frac{[H_2][CO_2]}{[H_2O][CO]}$$

A. $H_{2(g)} + CO_{2(g)} \leftrightharpoons H_2O_{(g)} + CO_{(g)}$
B. $H_2O_{(g)} + CO_{(g)} \leftrightharpoons H_{2(g)} + CO_{2(g)}$
C. $H_2O_{(g)} + CO_{(g)} \leftrightharpoons 2H^+_{(aq)} + CO_{2(g)}$
D. $2H^+_{(aq)} + CO_{2(g)} \leftrightharpoons H_2O_{(g)} + CO_{(g)}$

21. $2SO_{2(g)} + O_{2(g)} \leftrightharpoons 2SO_{3(g)}$

If the value of K_c in this reaction at 500 K is 5.0×10^{-2}, then the value of K_p for this reaction will be

(**Note**: R = 0.0821 dm^3 atm K^{-1} mol^{-1})

A. 1.2×10^{-3}
B. 5.0×10^{-2}
C. 2.9×10^{-5}
D. 7.2×10^{-7}

22. $N_2O_{4(g)} \leftrightharpoons 2NO_{2(g)}$ $\quad K_c = 1.09 \times 10^{-7}$ at 200°C

The given value of equilibrium constant indicates that N_2O_4 at 200°C shows

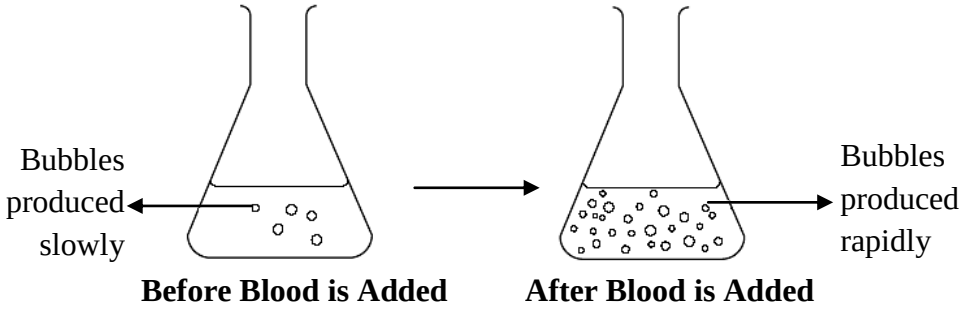
	Stability	**Rate of Decomposition**
A	high	rapid
B	low	rapid
C	high	slow
D	low	slow

23. A substance that delays the progress of a chemical reaction by decreasing effective collisions among reactant molecules is known as an

A. activator.
B. inhibitor.
C. auto-catalyst.
D. induced catalyst.

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24. A solution of hydrogen peroxide releases oxygen slowly at room temperature. When blood is added to the solution, its impact can be observed in the following diagram.



Bubbles are produced rapidly because blood

A. contains catalase enzyme.
B. supplies dissolved oxygen.
C. increases the concentration of hydrogen peroxide.
D. neutralises hydrogen peroxide with the release of oxygen.

25. If the rate of reaction is independent of the concentration of water, then the reaction is said to be of

A. zero order.
B. third order.
C. second order.
D. pseudo first order.

26. In cryoscopy, the formula for finding molar (molecular) mass of a solute is

A. $M_2 = \frac{\Delta T_f}{W_2}\frac{1000K_f}{W_1}$

B. $M_2 = \frac{\Delta T_f}{K_f}\frac{1000W_1}{W_2}$

C. $M_2 = \frac{K_f}{\Delta T_f}\frac{1000W_2}{W_1}$

D. $M_2 = \frac{K_f}{W_2}\frac{1000W_1}{\Delta T_f}$

Note:

M_2 = Molar mass of solute

ΔT_f = Depression in freezing point

K_f = Molal freezing point constant

W_1 = Mass of solvent in kg

W_2 = Mass of solute

27. A colloid is formed when water is mixed with

A. vinegar.
B. egg white.
C. lime water.
D. orange juice.

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28. ‘Dust particles present in smoke’ is a type of solution made up of

	Solute	Solvent
A	gas	gas
B	solid	liquid
C	solid	gas
D	gas	solid

29. A solution is prepared by adding 10 mg of NaCl in 500 g of water. The concentration of solution in parts per billion (ppb) would be

A. 2×10^{-7}
B. 2×10^{-4}
C. 2×10^{4}
D. 2×10^{7}

30. Enthalpy change in a chemical reaction is the same whether the reaction takes place in a single step or in multiple steps. This is known as

A. Hess’s law.
B. Graham’s law.
C. Le-Chatelier’s principle.
D. Heisenberg’s uncertainty principle.

31. All of the following statements describe lattice energy EXCEPT that it

A. can be determined indirectly through Born–Haber cycle.
B. helps to understand the structure, bonding and properties of ionic compounds.
C. is the amount of energy required to break one mole of a particular bond to form neutral atoms.
D. is the enthalpy of formation of one mole of the ionic compound from gaseous ions under standard conditions.

32. If 500 J heat is absorbed by a system and 250 J work is done on the system, then the change in internal energy (ΔE) of this system would be

A. –750
B. –250
C. +250
D. +750

33. Which of the following statements is FALSE about the working of bomb calorimeter?

A. Sample is placed in a crucible inside the bomb calorimeter.
B. The test substance is heated by passing the current through water.
C. Initial temperature is measured before the test substance is ignited.
D. Oxygen is provided through a valve until the pressure inside is about 20 atm.

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34. The oxidation number of sulphur in SO_3^{2-} is

A. + 2
B. + 3
C. + 4
D. + 6

35. Three elements X, Y and Z have standard reduction potential of +1.50, –2.87 and +0.771 volts respectively.

Which element(s) will lose electrons and undergo oxidation?

A. X only
B. Y only
C. X and Z
D. X and Y

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